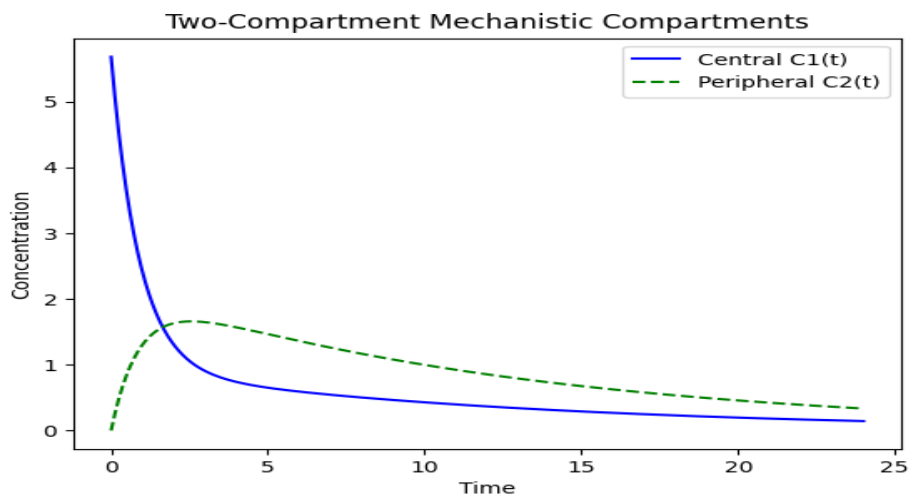


Note: Auto-demoted to two-compartment for Subject S1 ($\Delta AIC_c < 2$ and/or poor 3c rate separation / weak tail or $n < 8$).

Subject S1

Central vs. Peripheral Compartments



Study Metadata

Field	Value
study_id	UserStudy1
species	Mus musculus
route	IV bolus
dose	100
model	three
time_units	h
conc_units	a.u.
allow_demote	True
k10	0.35
k12	0.6
k21	0.25
k13	0.35
k31	0.12
V1	0.04

V2	0.06
V3	0.1
subject	S1

Data Summary

n	42
Time span (min → max)	0.00 → 24.00 h
C_max (observed)	5.66 a.u.
Global t½ (log-linear)	4.48 h
m_early / m_late (log-slopes)	1.72

Fit Results (with 95% CI)

Parameter	Estimate	95% CI
A	0.319	[0.0568, 0.582]
alpha	1.66	[-1.01, 4.33]
B	5.37	[5.11, 5.62]
beta	0.174	[0.166, 0.182]
Vd_ss	0.1	
Cl	0.014	

Goodness-of-Fit

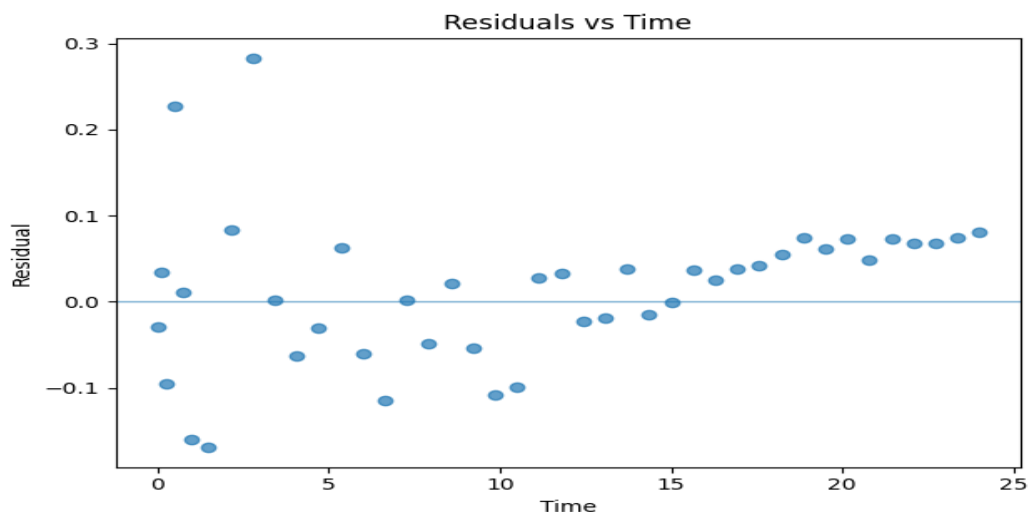
R ²	0.998
AIC	-197.7
AICc	-196.6

Model Selection Diagnostics

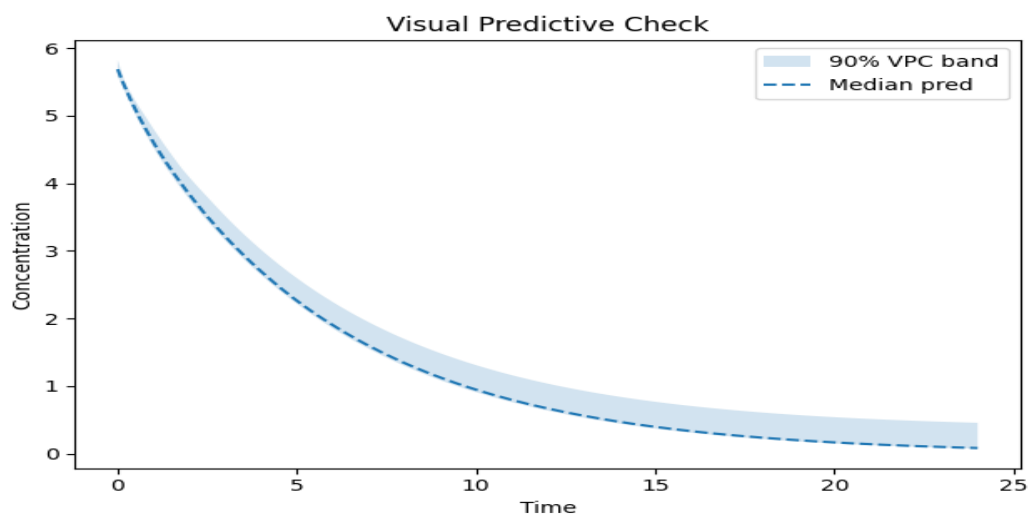
Criterion	AICc (linear)
AICc (2c)	-196.64
AICc (3c)	-195.23
ΔAICc (3c – 2c)	1.41

rate_sep_rel (3c)	0.007
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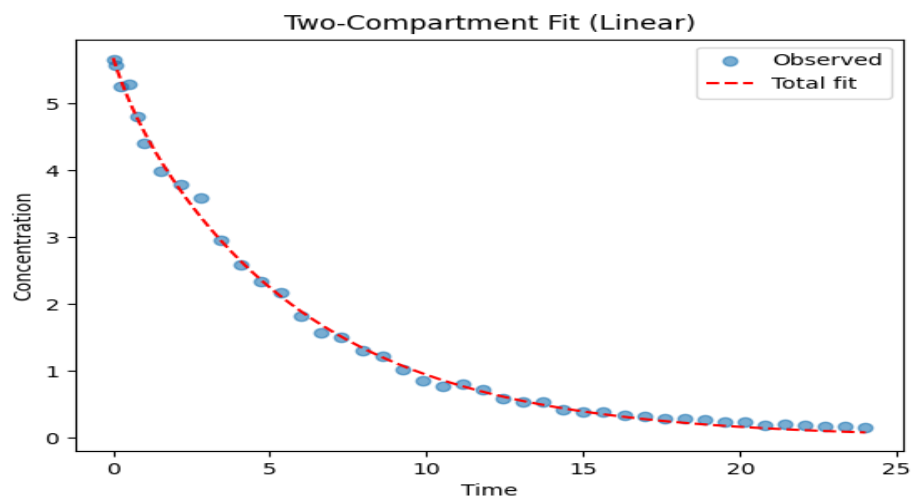
Residual Diagnostics



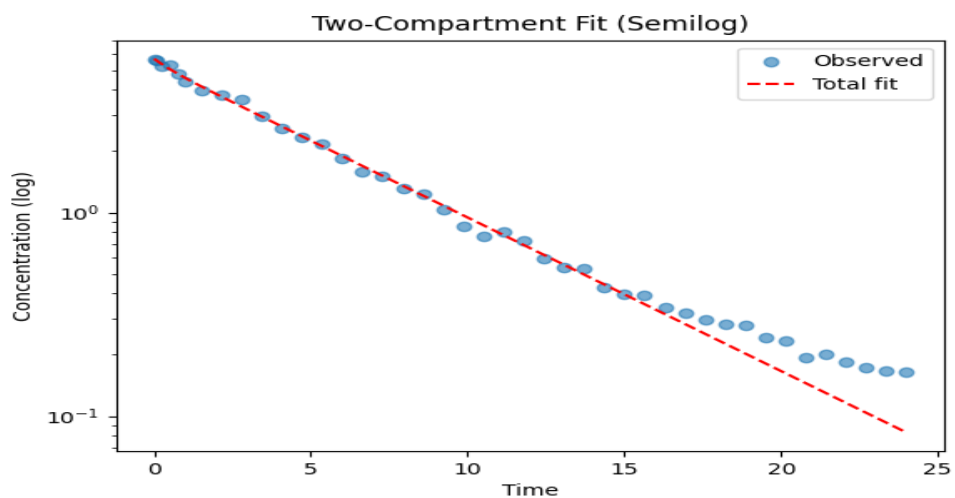
Visual Predictive Check (5–95%)



Linear Fit



Semilog Fit



Dosing Timeline

